Section 02

Industrial Security (Bachelor)

IS Security

Section 02 – ICS and SCADA Fundamentals

ICS and SCADA Fundamentals

Lecture Notes

Department of Computer Science

Outline – Section 02

Learning Objectives

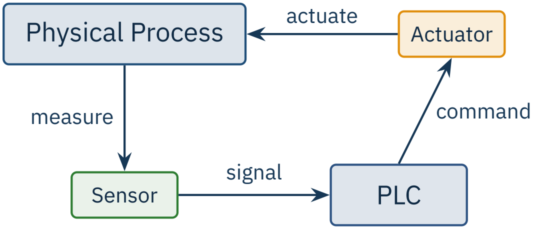
By the end of this section you will be able to

- Identify the components of a typical industrial control loop.
- Distinguish PLC, RTU, DCS, SCADA, HMI, Historian, and SIS.
- Explain the scan cycle and determinism requirements of PLCs.
- Read a simple ladder-logic program.
- Recognise why Safety Instrumented Systems must remain independent.

1

Control Loop and Field Devices

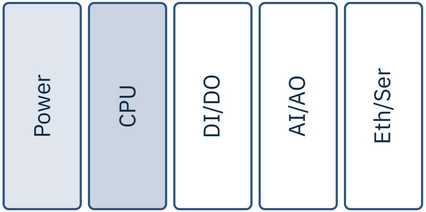
Anatomy of a Control Loop



Cycle:
10 μs – 100 ms
Deterministic

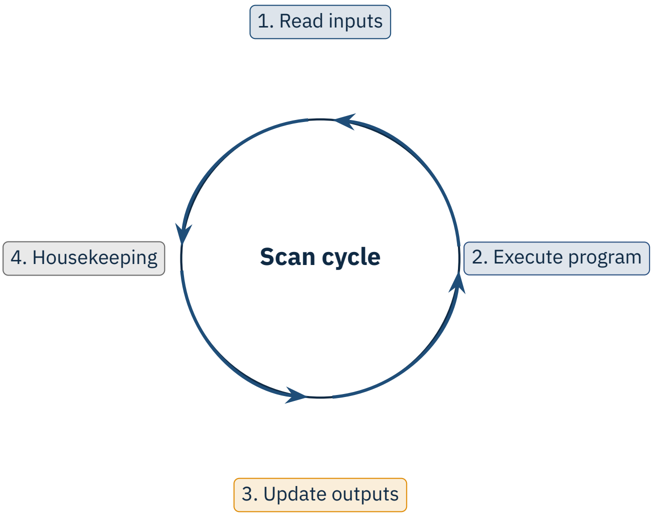
Programmable Logic Controllers (PLCs)

- Ruggedised industrial computer
- First model: Modicon 084 (1969)
- Replaced relay logic in plants
- IEC 61131-3 languages:
    - Ladder Diagram (LD)
    - Structured Text (ST)
    - Function Block (FBD)
    - Sequential Function Chart
    - Instruction List

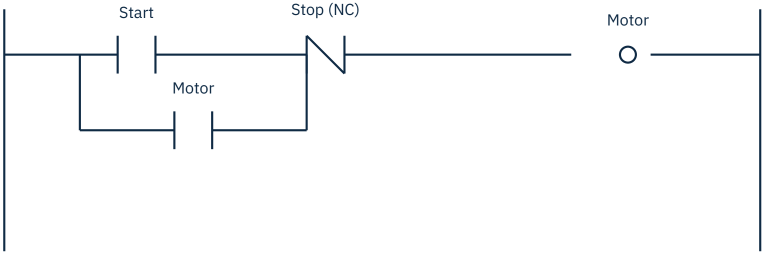


Source: D. Morley, “History of the PLC”, Modicon, 2003 | IEC 61131-3:2013.

PLC Scan Cycle



Ladder Logic — A Motor Start/Stop



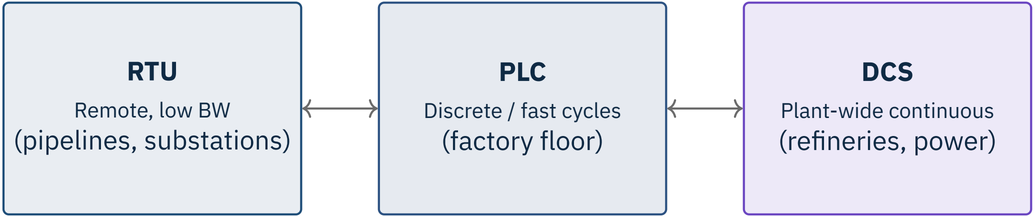
What it does

`Start` energises the `Motor` coil. The seal-in branch around `Start` latches the coil so it stays on. `Stop` (NC) breaks both paths.

2

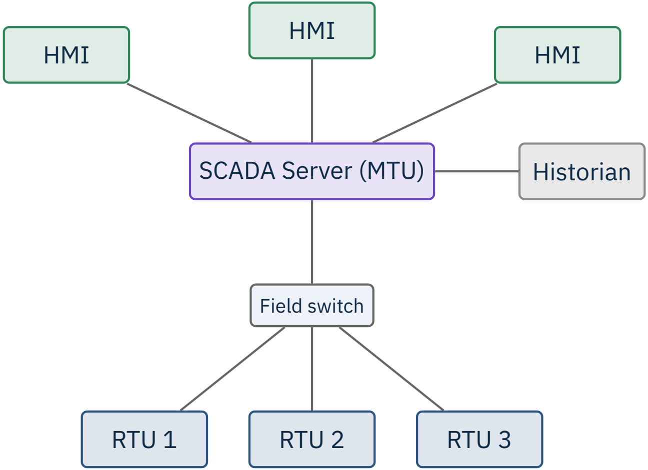
Plant-Wide Architectures

RTU vs. PLC vs. DCS



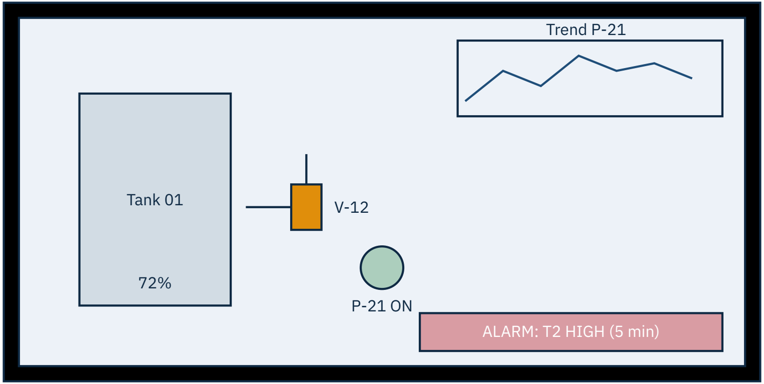
The boundary blurs in modern systems.

SCADA Architecture



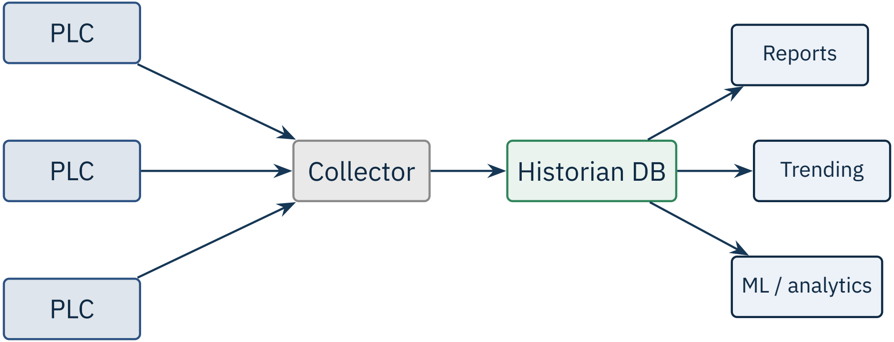
Source: Examples: Inductive Automation Ignition, Siemens WinCC OA, AVEVA System Platform, Schneider EcoStruxure Geo SCADA.

HMI – Human Machine Interface



Mimic diagrams, trends, alarm bars. Bad HMI design = accidents (Three Mile Island, 1979).

Historian – Time-Series of the Plant

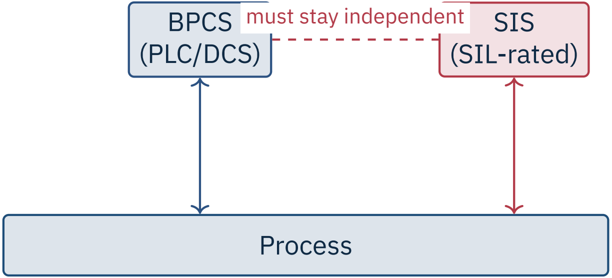


Historians bridge OT and IT; a frequent pivot point for attackers.

3

Safety Layer

Safety Instrumented Systems (SIS)



Safety layer

The SIS brings the process to a safe state on demand. **IEC 61511 / 61508**, SIL 1–4.

Source: IEC 61508 (2010, ed. 2.0), IEC 61511 (2016–17, ed. 2.0) | TRITON: Dragos *TRISIS Malware Analysis*, 2017.

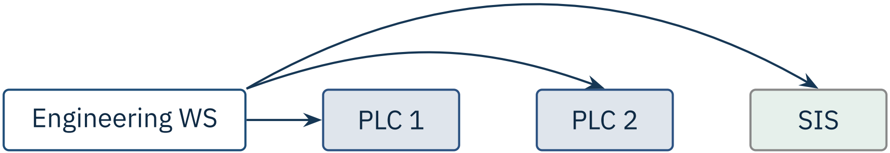
IEC 61131-3 – PLC Programming Languages

LD Ladder Diagram	**ST** Structured Text	**FBD** Function Block	**SFC** Sequential FC	**IL** Instruction List

The big two in the wild

LD dominates discrete manufacturing; ST is the cross-vendor portable choice for new development.

Engineering Workstation – High-Value Target



Caution

Why attackers love it The engineering WS holds the credentials, project files, and the ability to *re-flash logic*. Treat it as the crown jewel of the plant.

Process Types – Batch, Continuous, Discrete

Type	Examples	Control system flavour
Continuous	Refinery, power, water	DCS, tight feedback control
Batch	Pharma, food, chemicals	DCS + recipe management (ISA-88)
Discrete	Auto assembly, packaging	PLCs, fast cycle, ladder logic

Why this matters for security

Different processes have different "safe states" – shutdown is fine for discrete, may be catastrophic for continuous. IR plans must reflect that.

Summary – Section 02

★ Key Takeaway: What to remember from this section

- Control loops close from sensors through PLC/DCS to actuators.
- SCADA aggregates field data; HMIs deliver situational awareness.
- Historians are the long-term memory and a common IT/OT bridge.
- SIS must remain independent — attacking it was TRITON's goal in 2017.

References – Section 02

- IEC 61131-3:2013. *Programmable controllers – Part 3: Programming languages*.
- IEC 61508 (ed. 2.0, 2010). *Functional safety of E/E/PE safety-related systems*.
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- ISA. *ISA-101 Human Machine Interfaces for Process Automation Systems*, 2015.
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- Krutz, R.L. *Securing SCADA Systems*, Wiley, 2006.
- Macaulay, T., Singer, B. *Cybersecurity for Industrial Control Systems*, CRC Press, 2012.

End of Section 02

ICS and SCADA Fundamentals

Questions and discussion

IS Security